Individual Diagnostic Requirement Specification

CN200S ECM (1.8L)

Version: V1.0

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Requirement wording conventions

Within this document the following conventions are applied when describing specific compliance requirements:

The word “Should” denotes a preference or desired conformance.

The word “Shall” is used in the following ways:

1. To state a binding requirement on the special network, or the ECUs which comprise the network, which is verifiable by external manipulation and/or observation of an ECU or the network.

2. To state a binding requirement upon an ECU’s requirements document that is verifiable through a review of the document.

The word “Must” is used to state a binding requirement upon ECUs on the network which have a corresponding ECU requirements document. These requirements will be verified as part of the component verification.

NOTE: In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence.

NOTE: Nothing in this specification supersedes applicable laws and regulations unless a specific exemption has been obtained.

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History record of the document

The following table contains the revision history for this specification.
Changes will not be listed if corrections of the spelling or formal changes are made.

Version#	Date	Editor	Description of the revision
V1.0	2014-7-22	Liao wenqing	Initial release

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1 Introduction

1.1 Scope

The detailed description about this chapter can be found in chapter 1.1 of the document *CN200S General Diagnostic Requirement Specification*.

NOTE: For CN200S vehicle, ECM need to realize the software download function, the detailed requirements about software download can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

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2 Normative references

This document references ISO, SGMW and other publications as listed in the following sub sections.

2.1 ISO Publications

ISO 11898-1:2003 Road Vehicles - Controller Area Networks (CAN) - Part 1: Data link layer and physical signalling

ISO 11898-2:2003 Road Vehicles - Controller Area Networks (CAN) – Part 2: High-speed medium access unit

ISO 15765-1:2004 Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information

ISO 15765-2:2004 Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services

ISO 15765-3:2004 Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of unified diagnostic services (UDS on CAN)

ISO 14229-1:2006 Road Vehicles - Unified diagnostic services (UDS) - Specification and requirements

ISO 15031-6:2005 Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 6: Diagnostic trouble code definitions

2.2 SGMW Standard

CN200S General Diagnostic Requirement Specification
BT/SGMWJ 0858:2013 高速 CAN 物理层规范

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3 Terms, definitions and abbreviations

3.1 ISO Terms

This document makes use of terms defined in the ISO 15765-2, ISO 15765-3 and ISO 14229-1.

3.2 Specific Terms Defined By This Document

Table 3-1Specific Terms Defined By This Document

Abbreviation/缩写	Description/描述
msec	**m**illi**sec**ond
CANId	**CAN I**dentifier
ABS/ESP	**A**nti-lock **B**rake **S**ystem/**E**lectronic **S**tability **P**rogram
TPMS	**T**ire **P**ressure **M**onitor **S**ystem
SDM	**R**estraint **S**ensing and**D**iagnostic **M**odule
DCU	**D**omain **C**ontroller **U**nit
BCM	**B**ody **C**ontrol **M**odule
PEPS	**P**assive **E**ntry**&P**assive **S**tart
EPS	**E**lectronic **P**ower **S**teering
AC	**Auto fixed-temp AC**
IC	**I**nstrument **C**luster
ICE	**I**n **C**ar **E**ntertainment
ECM	**E**ngine **C**ontrol **M**odule
TCM	**T**ransmission **C**ontrol **M**odule

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4 General requirement

All the general requirements, with which the tester and ECM shall comply, can be found in section 4 of the document *CN200S General Diagnostic Requirement Specification*.

4.1 Diagnostic CAN identifiers definition

The following table provides an overview about the definition of ECM diagnostic CAN identifier.

Table 4-1ECM diagnostic CAN identifier

ECU		Functionally addressed request from tester to ECUs	Physically addressed request from tester to ECU	Response from ECU to tester
Name	Emission-related OBDECU (YES/NO)			
ECM	YES	0x7DF	0x7E0	0x7E8

4.2 ECU Diagnostic Address definition

The following table provides an overview about the definition of ECM diagnostic Address.

Table 4-2ECM Diagnostic Address

ECU Name	Description	Value	Comments
ECM	ECUDiagnosticAddress	0x11	ECM diagnostic address, which is defined in *CN200S General Diagnostic Requirement Specification*.

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5 Physical layer requirement

All the physical layer requirements, with which the tester and ECM shall comply, can be found in section 5 of the document *CN200S General Diagnostic Requirement Specification*.

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6 Data link layer requirement

All the data link layer requirements, with which the tester and ECM shall comply, can be found in section 6 of the document *CN200S General Diagnostic Requirement Specification*.

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7 Network layer requirement

All the network layer requirements, with which the tester and ECM shall comply, can be found in section 7 of the document *CN200S General Diagnostic Requirement Specification*.

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8 Application and session layers requirement

All the application and session layer requirements, with which the tester and ECM shall comply, can be found in section 8 of the document *CN200S General Diagnostic Requirement Specification*.

8.1 Diagnostic Service Table Overview

The table followed (Table 8-1) shows the services which are allowed during the defaultSession, programmingSession and extendedSession of ECM. ECMis allowed to support more other diagnostic services per session after confirmed by SGMW.

Table 8-1Diagnostic Service Table Overview

Service Name 服务名称	SID (Hex)	Sub-function/ 子功能 (Hex)	Session/会话						Security Access[1]/ 安全访问		
			Default 默认		Programming 编程		Extended 扩展		LV1 等级 1	LV 2 等级 2	LV 3 等级 3
			Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址			
Diagnostic and communication management functional unit/诊断与通信管理功能单元											
DiagnosticSessionControl	10	01	√[2]	√	√	√	√	√	×[2]	×	×
		02	√	×	√	×	√	×	×	×	×
		03	√	√	√	√	√	√	×	×	×
ECUReset	11	01	√	√	√	√	√	√	×	×	×
		02	√	√	×	×	√	√	×	×	×
SecurityAccess	27	01/02	×	×	×	×	√	×	×	×	×
		03/04	×	×	×	×	√	×	×	×	×
		07/08	×	×	√	×	×	×	×	×	×
CommunicationControl	28	00/03	×	×	×	×	√	√	×	×	×
TesterPresent	3E	00/80	√	√	√	√	√	√	×	×	×
ControlDTCSetting	85	01/02	×	×	×	×	√	√	×	×	×
ProgrammingMode	A5	01/03	×	×	×	×	√	√	×	×	×
Data transmission functional unit/数据传输功能单元											
ReadDataByIdentifier	22	-	√	√[3]	√	×	√	√[3]	×	×	×
DynamicallyDefineDataIdentifier	2C	-	√	×	×	×	√	×	×	×	×
WriteDataByIdentifier	2E	-	×	×	√	×	√	×	√[4]	×	×
Stored data transmission functional unit/存储数据传输功能单元											
ClearDiagnosticInformation	14	-	√	√	×	×	√	√	×	×	×
ReadDTCInformation	19	01/02/04/06/0A	√	√	×	×	√	√	×	×	×
InputOutput control functional unit/输入输出控制功能单元											
InputOutputControlByIdentifier	2F	-	×	×	×	×	√	×	×	×	×
Remote activation of routine functional unit/程序远程激活功能单元											

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Service Name 服务名称	SID (Hex)	Sub-function/ 子功能 (Hex)	Session/会话						Security Access[1]/ 安全访问		
			Default 默认		Programming 编程		Extended 扩展		LV1 等级 1	LV 2 等级 2	LV 3 等级 3
			Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址			
RoutineControl	31	01/03	×	×	√	×	√	×	×	√	√
Upload download functional unit/上传下载功能单元											
RequestDownload	34	-	×	×	√	×	×	×	×	×	√
TransferData	36	-	×	×	√	×	×	×	×	×	×
RequestTransferExit	37	-	×	×	√	×	×	×	×	×	×

NOTE:
注：

1. The definition of the security access level can be found in chapter 错误！未找到引用源。 of this specification.
有关安全访问等级的定义可参见本规范中的第错误！未找到引用源。章节。
2. '√': Support; '×': Not support, '-': Not Applicable.(N/A)
"√"：支持；"×"：不支持；"-"不适用。
3. If the ReadDataByIdentifier (22 hex) service request message is a single frame, the functional addressing method shall be supported; otherwise, the functional addressing method is not required to be supported.
如果通过标识符读数据（22 hex）服务请求消息为单帧，则应（shall）支持功能寻址，否则，不需要支持功能寻址。
4. All the WriteDataByIdentifier (2E hex) service in all ECUs shall need the security access, and the security level needed by this service depends on each DID(see annex C)
所有 ECU 的通过标识符写数据（2E hex）服务均应（shall）需要安全访问，该服务所需的安全等级取决于具体 DID（见附录 C）。
5. the RoutineControl (31 hex) service need the security access, the security level needed by this service depends on each RID which is defined in this document.
RoutineControl (31 hex) 服务所需的安全访问等级，在本文附录中的 RID 中具体定义。

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9 Diagnostic service documentation structure

All the requirements about diagnostic service documentation structure, with which the tester and ECM shall comply, can be found in section 9 of the document *CN200S General Diagnostic Requirement Specification*.

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10 Diagnostic and communication management functional unit

10.1 DiagnosticSessionControl (10 hex)

10.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.1.2 Request message

Table 10-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	DiagnosticSessionControl Request Service Id	10
#2	DiagnosticSessionType=[DefaultSession ProgrammingSession ExtendedDiagnosticSession]	 01 02 03

10.1.3 Positive response message

Table 10-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	DiagnosticSessionControl Response Service Id	50
#2	DiagnosticSessionType=[DefaultSession ProgrammingSession ExtendedDiagnosticSession]	 01 02 03
 #3 #4 #5 #6	sessionParameterRecord[]#1=[$P2_{CAN_ECU_max}$(high byte) $P2_{CAN_ECU_max}$(low byte) $P2^*_{CAN_ECU_max}$(high byte) $P2^*_{CAN_ECU_max}$(low byte)]	 00 32 01 F4

NOTE: The definition about parameter$P2_{CAN_ECU_max}$ and $P2^*_{CAN_ECU_max}$ can be found in chapter 8.2.2 of the document *CN200S General Diagnostic Requirement Specification* and ISO 15765-3. These two parameters in positive response message are the ones that are used by ECM internally, if the value of this two parameters which is used by ECM are different from the definition in this document, it should be determined by SGMW and ECM Supplier together.

10.1.4 Supported negative response codes

Table 10-3 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

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10.2 ECUReset (11 hex)

10.2.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.2.2 Request message

Table 10-4 Request message definition

A_Data Byte	Parameter name	Hex value
#1	ECUReset Request Service Id	11
#2	resetType =[hardReset keyOffOnReset]	 01 02

10.2.3 Positive response message

Table 10-5 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ECUReset Response Id	51
#2	resetType =[hardReset keyOffOnReset]	 01 02

10.2.4 Supported negative response codes

Table 10-6 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ECM (1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

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10.3 SecurityAccess (27 hex)

10.3.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.3.2 Request message

Table 10-7 Request message definition

A_DataByte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType=requestSeed	01/03/07

Table 10-8 Request message definition

A_DataByte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType =sendKey	02/04/08
	securityKey[]=[	
#3	key #1(MSB)	00-FF
#4	key #2	:
#5	key #3	:
#6	key#4(LSB)]	00-FF

10.3.3 Positive response message

Table 10-9 Positive response message definition (sub-function = requestSeed)

A_DataByte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType=requestSeed	01/03/07
	securitySeed[]=[	
#3	seed#1(MSB)	00-FF
#4	seed#2	:
#5	seed#3	:
#6	seed#4(LSB)]	00-FF

Table 10-10 Positive response message definition (sub-function = sendKey)

A_DataByte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType =sendKey	02/04/08

10.3.4 Supported negative response codes

Table 10-11 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

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24	**requestSequenceError**
35	**invalidKey**
36	**exceededNumberOfAttempts**
37	**requireTimeDelayNotExpired**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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10.4 CommunicationControl (28 hex)

10.5 TesterPresent (3E hex)

10.5.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.5.2 Request message

Table 10-12 Request message definition

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Request Service Id	3E
#2	zeroSubFunction=[zeroSubFunction (suppressPosRspMsgIndicationBit =0) zeroSubFunction (suppressPosRspMsgIndicationBit =1)]	 00 80

10.5.3 Positive response message

Table 10-13 Positive response message definition(sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =0))

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Response Service Id	7E
#2	zeroSubFunction= zeroSubFunction (suppressPosRspMsgIndicationBit =0)	00

sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of suppressPosRspMsgIndicationBit =1.

10.5.4 Supported negative response codes

Table 10-14 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

10.6 AccessTimingParameter (83 hex)

N/A

10.7 SecuredDataTransmission (84 hex)

N/A

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10.8 ControlDTCSetting (85 hex)

10.9 ResponseOnEvent (86 hex)

N/A

10.10 LinkControl (87 hex)

N/A

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10.11 ProgrammingMode (A5 hex)

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11 Data transmission functional unit

11.1 ReadDataByIdentifier (22 hex)

11.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

NOTE: The maximum number of dataIdentifier that can be simultaneously requested is fifteen.

NOTE: The DID which used to identify the general data record for all the ECUs called vehicle-manufacturer-specific DID, the definition about vehicle-manufacturer-specific DID can be found in chapter C.1 of Annex C. The DID which used to identify the individual data record for each ECU called ECU-supplier-specific DID, the definition about ECU-supplier-specific DID can be found in in chapter C.2 of Annex C.

11.1.2 Request message

Table 11-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Request Service Id	22
	dataIdentifier[]#1=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
:	:	:
	dataIdentifier[]#m[a]=[	
#n-1	byte #1(MSB)	00-FF
#n	byte #2(LSB)]	00-FF
a. m≤15		

11.1.3 Positive response message

Table 11-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Response Service Id	62
	dataIdentifier[]#1=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
	dataRecord[]#1=[	
#4	data #1	00-FF
:	:	:
#(k-1)+4	data #k[a]]	00-FF
:	:	:
	dataIdentifier[]#m[b]=[	
#n-(o-1)-2	byte #1(MSB)	00-FF
#n-(o-1)-1	byte #2(LSB)]	00-FF
	dataRecord[]#m[b]=[	

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#n-(o-1) : #n	data #1 : data #o[a]]	00-FF : 00-FF
a. The number of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C. b. m≤15		

11.1.4 Supported negative response codes

Table 11-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

11.2 ReadMemoryByAddress (23 hex)

N/A

11.3 ReadScalingDataByIdentifier (24 hex)

N/A

11.4 ReadDataByPeriodicIdentifier (2A hex)

N/A

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11.5 DynamicallyDefineDataIdentifier (2C hex)

11.6 WriteDataByIdentifier (2E hex)

11.6.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

11.6.2 Request message

Table 11-4 Request message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Request Service Id	2E
	dataIdentifier[]=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
	dataRecord[]=[	
#4	data #1	00-FF
:	:	:
#m+3	data #m[a]]	00-FF

a. The content of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C.

11.6.3 Positive response message

Table 11-5 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Response Service Id	6E
	dataIdentifier[]=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF

11.6.4 Supported negative response codes

Table 11-6 Supported negative response codes

Hex	Description
33	**securityAccessDenied**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**
72	**generalProgrammingFailure**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

11.7 WriteMemoryByAddress (3D hex)

N/A

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12 Stored data transmission functional unit

12.1 ClearDiagnosticInformation (14 hex)

12.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

12.1.2 Request message

Table 12-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Request Service Id	14
	groupOfDTC[]=[	
#2	groupOfDTCHighByte	FF
#3	groupOfDTCMiddleByte	FF
#4	groupOfDTCLowByte]	FF

12.1.3 Positive response message

Table 12-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Response Service Id	54

12.1.4 Supported negative response codes

Table 12-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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12.2 ReadDTCInformation (19 hex)

12.2.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

12.2.2 Request message

Table 12-4 Request message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=[reportNumberOfDTCByStatusMask reportDTCByStatusMask]	 01 02
#3	DTCStatusMask	00-FF

Table 12-5 Request message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
 #3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCSnapshotRecordNumber	FF[a]
a. A value of FF hex requests the ECU to report all stored DTCSnapshot data records at once.		

Table 12-6 Request message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
 #3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCExtendedDataRecordNumber	00-FF[a]
a. A value of FF hex requests the ECU to report all stored DTCExtendedData records in a single response message.		

Table 12-7 Request message definition (Sub-function = reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportSupportedDTC	0A

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1.1.1 Positive response message

Table 12-8 Positive response message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportNumberOfDTCByStatusMask	01
#3	DTCStatusAvailabilityMask	FF
#4	ISO 15031-6 DTCFormat	00
	DTCCount[]=[	
#5	DTCCountHighByte	00-FF
#6	DTCCountLowByte]	00-FF

Table 12-9 Positive response message definition (sub-function = reportDTCByStatusMask, reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
	reportType=[	
#2	reportDTCByStatusMask	02
	reportSupportedDTC]	0A
#3	DTCStatusAvailabilityMask	FF
	DTCAndStatusRecord[] = [	
#4	DTCHighByte #1	00-FF
#5	DTCMiddleByte #1	00-FF
#6	DTCLowByte #1	00-FF
#7	statusOfDTC #1	00-FF
#8	DTCHighByte #2	00-FF
#9	DTCMiddleByte #2	00-FF
#10	DTCLowByte #2	00-FF
#11	statusOfDTC #2	00-FF
:	:	:
#n-3	DTCHighByte #m	00-FF
#n-2	DTCMiddleByte #m	00-FF
#n-1	DTCLowByte #m	00-FF
#n	statusOfDTC #m]	00-FF

NOTE: If reportType = reportSupportedDTC:

1. ECM should report the status of all the DTC's which is supported by ECM.
2. ECM should report the status at the time of receiving the requirement.

Table 12-10 Positive response message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
	DTCAndStatusRecord[]=[	
#3	DTCHighByte	00-FF
#4	DTCMiddleByte	00-FF
#5	DTCLowByte	00-FF
#6	statusOfDTC]	00-FF
#7	DTCSnapshotRecordNumber#1	11
#8	DTCSnapshotRecordNumberOfIdentifiers#1	10

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	DTCSnapshotRecord[]#1[a]=[	
#9	Calculated load value DataIdentifier(MSB)	00
#10	Calculated load value DataIdentifier(LSB)	04
#11	Calculated load value	00-FF
#12	Engine Coolant temperature DataIdentifier(MSB)	00
#13	Engine Coolant temperature DataIdentifier(LSB)	05
#14	Engine Coolant temperature	00-FF
#15	Engine RPM DataIdentifier (MSB)	00
#16	Engine RPM DataIdentifier (LSB)	0C
#17	Engine RPM (High Byte)	00-FF
#18	Engine RPM (Low Byte)	00-FF
#19	Fuel System Staus DataIdentifier (MSB)	00
#20	Fuel System Staus DataIdentifier(LSB)	03
#21	Fuel System Staus (High Byte)	00-FF
#22	Fuel System Staus (Low Byte)	00-FF
#23	Short Term Fuel Trim bank1 DataIdentifier (MSB)	00
#24	Short Term Fuel Trim bank1 DataIdentifier (LSB)	06
#25	Short Term Fuel Trim bank1	00-FF
#26	Long Term Fuel Trim bank1 DataIdentifier (MSB)	00
#27	Long Term Fuel Trim bank1 DataIdentifier (LSB)	07
#28	Long Term Fuel Trim bank1	00-FF
#29	Manifold pressure (absolute) DataIdentifier (MSB)	00
#30	Manifold pressure (absolute) DataIdentifier (LSB)	0B
#31	Manifold pressure (absolute)	00-FF
#32	Vehicle speed Sensor DataIdentifier (MSB)	00
#33	Vehicle speed Sensor DataIdentifier (MSB)	0D
#34	Vehicle speed Sensor	00-FF
#35	Ignition timing advance for #1 DataIdentifier (MSB)	00
#36	Ignition timing advance for #1 DataIdentifier (LSB)	0E
#37	Ignition timing advance for #1	00-FF
#38	Intake air temperature DataIdentifier (MSB)	00
#39	Intake air temperature DataIdentifier (LSB)	0F
#40	Intake air temperature	00-FF
#41	Absolute throttle position DataIdentifier (MSB)	00
#42	Absolute throttle position DataIdentifier (LSB)	11
#43	Absolute throttle position	00-FF
#44	Commanded Evaporative Purge DataIdentifier (MSB)	00
#45	Commanded Evaporative Purge DataIdentifier (LSB)	2E
#46	Commanded Evaporative Purge	00-FF
#47	Relative Throttle Position DataIdentifier (MSB)	00
#48	Relative Throttle Position DataIdentifier (LSB)	45
#49	Relative Throttle Position	00-FF
#50	Ambient air temperature DataIdentifier (MSB)	00
#51	Ambient air temperature DataIdentifier (LSB)	46
#52	Ambient air temperature	00-FF
#53	Absolute Throttle Position B DataIdentifier (MSB)	00
#54	Absolute Throttle Position B DataIdentifier (MSB)	47
#55	Absolute Throttle Position B	00-FF
#56	Commanded Throttle Actuator Control DataIdentifier (MSB)	00
#57	Commanded Throttle Actuator Control DataIdentifier (LSB)	4C

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#58	Commanded Throttle Actuator Control	00-FF
a. The detailed definition can be found in the following Tables.		

Table 12-11 Positive response message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
	DTCAndStatusRecord[]=[	
#3	DTCHighByte	00-FF
#4	DTCMiddleByte	00-FF
#5	DTCLowByte	00-FF
#6	statusOfDTC]	00-FF
#7	DTCExtendedDataRecordNumber #1[a]	01
#8	Fault occurrence counter[a]	00-FF
#9	DTCExtendedDataRecordNumber #2[a]	02
#10	Fault pending counter[a]	00-FF
#11	DTCExtendedDataRecordNumber #3[a]	03
#12	DTC Aging counter[a]	00-FF
#13	DTCExtendedDataRecordNumber #4[a]	04
#14	DTC Aged counter[a]	00-FF
a. This part is the DTCExtendedDataRecord for all ECUs, detailed definitions of DTC extended data record can be found in chapter 12.2.3.2 of *CN200S General Diagnostic Requirement Specification*. Whether the parameter appeared or not and the number of these parameters that appeared depend on the specific service requested.		

1.1.2 Supported negative response codes

Table 12-12 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ECM(1.8L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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13 InputOutput control functional unit

13.1 InputOutputControlByIdentifier (2F hex)

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14 Remote activation of routine functional unit

14.1 RoutineControl (31 hex)

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15 Upload download functional unit

15.1 RequestDownload (34 hex)

15.2 RequestUpload (35 hex)

N/A

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15.3 TransferData (36 hex)

15.4 RequestTransferExit (37 hex)

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16 ECU programming requirements and process

The detailed description about this chapter can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

NOTE: For CN200S vehicle, ECM need to realize the software download function, the detailed requirements about software download can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

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Annex A (normative) Global parameter definitions

The detailed description about this Annex can be found in Annex A of the document *CN200S General Diagnostic Requirement Specification*.

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Annex B (normative) Diagnostic and communication management functional unit data parameter definitions

The detailed description about this Annex can be found in Annex B of the document *CN200S General Diagnostic Requirement Specification*.

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Annex C (normative) Data transmission functional unit data parameter definitions

C.1 Vehicle-manufacturer-specific DID definitions

The following table lists the detailed definition about vehicle-manufacturer-specific DID. The function and actual meaning about these DIDs can be found in the document of *CN200S General Diagnostic Requirement Specification*.

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C.2 ECU-supplier-specific DID definitions

The following table lists the definition about ECU-supplier-specific DID. According to the requirements of the document *CN200S General Diagnostic Requirement Specification*, the range which ECM used is 0x0101~0x0FFF.

DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0200	maximum attainable indicated torque	√	x	2	Hex
0201	desired indicated engine torque	√	x	2	Hex
0203	engine temperature	√	x	1	Hex
0204	engine speed	√	x	2	Hex
0205	time counter at end of start	√	x	2	Hex
0206	fuel level	√	x	2	Hex
0209	ambient pressure	√	x	2	Hex
020B	throttle angle with respect to lower mechanical stop	√	x	2	Hex
020D	intake air temperature	√	x	1	Hex
020E	ambient air temperature	√	x	1	Hex
020F	pressure upstream throttle valve	√	x	2	Hex
0210	normalized angle acceleration pedal	√	x	2	Hex
0211	volumetric flow of fuel in L/s	√	x	2	Hex
0212	desired pressure upstream throttle valve	√	x	2	Hex
0214	exhaust gas temperature at manifold out of model	√	x	2	Hex
021B	air-mass flow HFM	√	x	2	Hex
021C	ISO15031_PID03a	√	x	1	Hex
021D	ISO15031_PID03b	√	x	1	Hex
021E	relative air mass (calc. load value) acc. To SAE J1979	√	x	1	Hex
0220	Air mass flow	√	x	1	Hex
0223	output voltage oxygen sensor downstream catalyst	√	x	1	Hex
0224	output voltage oxygen sensor downstream catalyst bank 2	√	x	1	Hex
0225	relative air charge	√	x	1	Hex
0227	ETS-path as environmental condition for diagnosis of function monitoring	√	x	1	Hex
0228	Torque path in function and function monitoring as environmental cond. of diag	√	x	1	Hex
0229	actual torque at response of torque comparison in the function monitoring	√	x	1	Hex
022B	model-based substitute value for engine temperature signal in case of error	√	x	1	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
022C	desired throttle angle w.r.t. to lower mechanical stop	√	x	1	Hex
022D	exhaust temperature downstream catalyst (model)	√	x	1	Hex
022F	Engine coolant temperature, linearised and calculated	√	x	1	Hex
0230	model-based reference value for plausibility check of engine temperature signal	√	x	2	Hex
0231	Calculated actual torque in function monitoring	√	x	1	Hex
0232	Engaged gear	√	x	1	Hex
0234	relative fuel mass for storage (service tool)	√	x	2	Hex
0236	real ignition angle	√	x	1	Hex
0237	correction factor for mass-flow substitute load signal	√	x	2	Hex
023A	throttle blade position dependent on poti 1	√	x	2	Hex
023B	throttle blade position dependent on poti 2	√	x	2	Hex
023C	DLR for DV-E: Sum of the PID-components (Throttle actuator Control PWM)	√	x	2	Hex
023D	Diagnosis: 8 bit copy of wdknlp_w as ambient value	√	x	1	Hex
023F	Voltage DK-Poti 1 at the NLP	√	x	2	Hex
0240	sensor voltage poti 1 of throttle actuator at (lower) mechanical stop	√	x	1	Hex
0241	Automobile acceleration in the longitudinal direction	√	x	1	Hex
0243	signal voltage, ambient pressure sensor	√	x	2	Hex
0244	Ambient pressure without substitution on sensor error	√	x	2	Hex
0245	intake manifold pressure (absolute)	√	x	2	Hex
0246	Number of clutch operations detected	√	x	1	Hex
0247	intake air temperature, linearised and calculated	√	x	1	Hex
0248	vehicle speed output value to scan tool	√	x	2	Hex
0249	battery voltage; scanned value of ADC	√	x	2	Hex
0252	Angle: intake valve opens related to LWOT	√	x	2	Hex
0254	desired camshaft angle inlet valve opened	√	x	2	Hex
0256	oil temperature	√	x	2	Hex
0257	lambda controller output	√	x	2	Hex
0258	multiplicative correction of the mixture adaptation	√	x	2	Hex
0259	Lambda controller output bank 2	√	x	2	Hex
025A	multiplicative correction of the mixture adaptation bank 2	√	x	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
025B	sensor voltage from throttle potentiometer 1	√	x	2	Hex
025C	sensor voltage from throttle potentiometer 2	√	x	2	Hex
0260	additive correction of the mixture adaptation	√	x	2	Hex
0262	Voltage PWG potentiometer 1	√	x	2	Hex
0263	Voltage PWG potentiometer 2	√	x	2	Hex
0264	Doubled PWG potentiometer-2 voltage	√	x	2	Hex
0265	Actual value internal res. Ri-Nernst cell lambda sensor downstream cat	√	x	2	Hex
0266	Exhaust gas temperature in front of precatalyst out of model	√	x	2	Hex
0267	HFM air mass value	√	x	2	Hex
0268	Transfer function (Mrad/Mkurbelwelle) from transmission control	√	x	1	Hex
026B	Interface for mode $01+$02 short term fuel trim bank 1 for PID $06	√	x	1	Hex
026C	Interface for mode $01+$02 long term fuel trim bank 1 for PID $07	√	x	1	Hex
026D	injection time	√	x	4	Hex
026E	effective steady-state idle speed	√	x	1	Hex
0278	Location of oxygen sensors	√	x	1	Hex
027A	OBD requirements to which vehicle is designed	√	x	1	Hex
027B	counter for distance travelled while MIL is activated	√	x	2	Hex
027C	engine temperature at start	√	x	1	Hex
027D	standardized mass flow from canister purge control	√	x	2	Hex
027E	basic ignition angle	√	x	1	Hex
027F	fault counter, summary, counts emission relevant misfirings of all cylinders	√	x	2	Hex
0284	area of misfire, minimum engine speed	√	x	2	Hex
0285	area of misfire, maximum engine speed	√	x	2	Hex
0286	area of misfirings, minimum load	√	x	1	Hex
0287	area of misfirings, maximum load	√	x	1	Hex
0288	Cycle time forced lambda modulation	√	x	2	Hex
0289	desired torque change from the idle speed control (PD-part)	√	x	2	Hex
02A2	Actual intake manifold pressure sensor voltage	√	x	2	Hex
02AA	O_2 sensor voltage bank1, sensor 1	√	x	2	Hex
02AB	O_2 sensor voltage bank1, sensor 2	√	x	2	Hex
0311	ignition-timing retardation with retardation for dynamics, Cylinder #1	√	x	1	Hex
0312	ignition-timing retardation with retardation for dynamics, Cylinder #2	√	x	1	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0313	ignition-timing retardation with retardation for dynamics, Cylinder #3	√	x	1	Hex
0314	ignition-timing retardation with retardation for dynamics, Cylinder #4	√	x	1	Hex
0321	reference level knock control, 16bit, Cylinder #1	√	x	2	Hex
0322	reference level knock control, 16bit, Cylinder #2	√	x	2	Hex
0323	reference level knock control, 16bit, Cylinder #3	√	x	2	Hex
0324	reference level knock control, 16bit, Cylinder #4	√	x	2	Hex
0331	number of misfire: cylinder#1	√	x	2	Hex
0332	number of misfire: cylinder#2	√	x	2	Hex
0333	number of misfire: cylinder#3	√	x	2	Hex
0334	number of misfire: cylinder#4	√	x	2	Hex
0335	Cycle time forced lambda modulation	√	x	2	Hex
0D00	battery voltage	√	x	2	Hex
0D01	vehicle speed	√	x	2	Hex
0D05	indicated real engine torque	√	x	2	Hex
0D06	Drivingdistance since prime start	√	x	4	Hex
0F00	Coolant temperature of engine misses	√	x	1	Hex
0F01	Duty cycle for carbon	√	x	1	Hex
0F02	Intake VVT control PWM	√	x	1	Hex
0F03	Upstream O_2 heater status	√	x	1	Hex
0F04	Downstream O_2 heater status	√	x	1	Hex
0F05	Injection time cylinder#1	√	x	2	Hex
0F06	Injection time cylinder#2	√	x	2	Hex
0F07	Injection time cylinder#3	√	x	2	Hex
0F08	Injection time cylinder#4	√	x	2	Hex
0F09	λ closed loop control status	√	x	1	Hex
0F0A	Fuel pump relay status	√	x	1	Hex
0F0B	Main relay status	√	x	1	Hex
0F0C	Cooling fan 2 switch status	√	x	1	Hex
0F0D	Cooling fan 1 switch status	√	x	1	Hex
0F0E	A/C compressor clutch relay status	√	x	1	Hex
0F0F	A/C request switch status	√	x	1	Hex
0F10	A/C evaporator pressure	√	x	1	Hex
0F11	A/C evaporator temperature	√	x	1	Hex
0F12	Break test switch status	√	x	1	Hex
0F13	Break lamp switch status	√	x	1	Hex
0F14	Clutch switch status	√	x	1	Hex
0F15	MIL indicator	√	x	1	Hex
0F16	Cruise status	√	x	1	Hex
0F17	Cruise speed	√	x	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
NOTE: " √ ": Support without security access; " √ ": Support with security access (A superscript number 'n' after ' √ ' indicates that the number of the security level); "X": Not Supported.					

C.2.1 DID 0x0200-maximum attainable indicated torque

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
maximum attainable indicated torque	2	%	0.001525879	0	0	99.99847

C.2.2 DID 0x0201-desired indicated engine torque

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
desired indicated engine torque	2	%	0.001525879	0	0	99.99847

C.2.3 DID 0x0203-engine temperature

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
engine temperature	1	℃	0.75	-48	-48	143.25

C.2.4 DID 0x0204-engine speed

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
engine speed	2	rpm	0.25	0	0	16383.75

C.2.5 DID 0x0205-time counter at end of start

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
time counter at end of start	2	s	0.1	0	0	6553.5

C.2.6 DID 0x0206-fuel level

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
fuel level	2	L	0.1	0	0	6553.5

C.2.7 DID 0x0209-ambient pressure

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ambient pressure	2	hPa	0.0390625	0	0	2559.960938

C.2.8 DID 0x020B-throttle angle with respect to lower mechanical stop

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
throttle angle with respect to lower mechanical stop	2	%DK	0.024414063	-800	-800	799.9755859

C.2.9 DID 0x020D-intake air temperature

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
intake air temperature	1	℃	0.75	-48	-48	143.25

C.2.10 DID 0x020E-ambient air temperature

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ambient air temperature	1	℃	0.75	-48	-48	143.25

C.2.11 DID 0x020F-pressure upstream throttle valve

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
pressure upstream throttle valve	2	hPa	0.78125	0	0	51199.21875

C.2.12 DID 0x0210-normalized angle acceleration pedal

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
normalized angle acceleration pedal	2	%PED	0.001525879	0	0	99.99847

C.2.13 DID 0x0211-volumetric flow of fuel in L/s

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
volumetric flow of fuel in L/s	2	L/s	5.9605E-07	0	0	0.039061904

~~C.2.14 DID 0x0212-desired pressure upstream throttle valve~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~desired pressure upstream throttle valve~~	~~2~~	~~hPa~~	~~0.78125~~	~~0~~	~~0~~	~~51199.21875~~

~~C.2.15 DID 0x0214-exhaust gas temperature at manifold out of model~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~exhaust gas temperature at manifold out of model~~	~~2~~	~~℃~~	~~0.0234375~~	~~-273.15~~	~~-273.15~~	~~1262.826563~~

~~C.2.16 DID 0x021B-air-mass flow HFM~~

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~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~air-mass flow HFM~~	~~2~~	~~Kg/h~~	~~0.1~~	~~0~~	~~0~~	~~6553.5~~

C.2.17 DID 0x021C-air-ISO15031_PID03a

Data Name	#of bytes	Description
ISO15031_PID03a	1	Refer to the definition of “PID03a” in the document of 15031-5.

~~C.2.18 DID 0x021D-air-ISO15031_PID03b~~

~~Data Name~~	~~#of bytes~~	~~Description~~
~~ISO15031_PID03b~~	~~1~~	~~Refer to the definition of “PID03b” in the document of 15031-5.~~

C.2.19 DID 0x021E-relative air mass (calc. load value) acc. To SAE J1979

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
relative air mass (calc. load value) acc. To SAE J1979	1	%	0.390625	0	0	99.609375

C.2.20 DID 0x0220-Air mass flow

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Air mass flow	1	Kg/h	4	0	0	1020

C.2.21 DID 0x0223-output voltage oxygen sensor downstream catalyst

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
output voltage oxygen sensor downstream catalyst	1	V	0.005228758	-0.2	-0.2	1.1333

~~C.2.22 DID 0x0224-output voltage oxygen sensor downstream catalyst bank 2~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~output voltage oxygen sensor downstream catalyst bank 2~~	~~1~~	~~V~~	~~0.005228758~~	~~-0.2~~	~~-0.2~~	~~1.1333~~

C.2.23 DID 0x0225-relative air charge

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
relative air charge	1	%	0.75	0	0	191.25

C.2.24 DID 0x0227-ETS-path as environmental condition for diagnosis of function monitoring

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ETS-path as environmental condition for diagnosis of function monitoring	1	–	1	0	0	255

C.2.25 DID 0x0228-Torque path in function and function monitoring as environmental cond. of diag

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Torque path in function and function monitoring as environmental cond. of diag	1	–	1	0	0	255

C.2.26 DID 0x0229-actual torque at response of torque comparison in the function monitoring

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
actual torque at response of torque comparison in the function monitoring	1	%	0.390625	0	0	99.609375

C.2.27 DID 0x022B-model-based substitute value for engine temperature signal in case of error

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
model-based substitute value for engine temperature signal in case of error	1	℃	0.75	-48	-48	143.25

C.2.28 DID 0x022C-desired throttle angle w.r.t. to lower mechanical stop

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
desired throttle angle w.r.t. to lower mechanical stop	1	%DK	0.392156863	0	0	100

C.2.29 DID 0x022D-exhaust temperature downstream catalyst (model)

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
exhaust temperature downstream catalyst (model)	1	℃	5	-50	-50	1225

C.2.30 DID 0x022F-Engine coolant temperature, linearised and calculated

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Engine coolant temperature, linearised and calculated	1	℃	0.75	-48	-48	143.25

C.2.31 DID 0x0230-model-based reference value for plausibility check of engine temperature signal

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
model-based reference value for plausibility check of engine temperature signal	2	℃	0.75	-48	-48	49103.25

C.2.32 DID 0x0231-Calculated actual torque in function monitoring

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Calculated actual torque in function monitoring	1	%	0.390625	0	0	99.609375

C.2.33 DID 0x0232-Engaged gear

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Engaged gear	1	–	1	0	0	255

C.2.34 DID 0x0234-relative fuel mass for storage (service tool)

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
relative fuel mass for storage (service tool)	2	%	3	0	0	196605

C.2.35 DID 0x0236-real ignition angle

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
real ignition angle	1	Grad KW	0.75	-96	-96	95.25

C.2.36 DID 0x0237-correction factor for mass-flow substitute load signal

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
correction factor for mass-flow substitute load signal	2	–	6.1035E-05	0	0	3.999938965

C.2.37 DID 0x023A-throttle blade position dependent on poti 1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
throttle blade position dependent on poti 1	2	%DK	0.024414063	-800	-800	799.9755859

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C.2.38 DID 0x023B-throttle blade position dependent on poti 2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
throttle blade position dependent on poti 2	2	%DK	0.024414063	-800	-800	799.9755859

C.2.39 DID 0x023C-DLR for DV-E: Sum of the PID-components (Throttle actuator Control PWM)

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
DLR for DV-E: Sum of the PID-components (Throttle actuator Control PWM)	2	%	0.001525879	0	0	99.99847

C.2.40 DID 0x023D-Diagnosis: 8 bit copy of wdknlp_w as ambient value

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Diagnosis: 8 bit copy of wdknlp_w as ambient value	1	%DK	0.392156863	0	0	100

C.2.41 DID 0x023F-Voltage DK-Poti 1 at the NLP

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Voltage DK-Poti 1 at the NLP	2	V	0.001220703	0	0	79.9987793

C.2.42 DID 0x0240-sensor voltage poti 1 of throttle actuator at (lower) mechanical stop

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
sensor voltage poti 1 of throttle actuator at (lower) mechanical stop	1	V	0.01953125	0	0	4.98046875

C.2.43 DID 0x0241-Automobile acceleration in the longitudinal direction

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Automobile acceleration in the longitudinal direction	1	m/s^2	0.217	-27.776	-27.776	27.559

~~C.2.44 DID 0x0243-signal voltage, ambient pressure sensor~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~signal voltage, ambient pressure sensor~~	~~2~~	~~V~~	~~0.0048828125~~	~~0~~	~~0~~	~~319.995~~

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~~C.2.45 DID 0x0244-Ambient pressure without substitution on sensor error~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~Ambient pressure without substitution on sensor error~~	~~2~~	~~hPa~~	~~0.0390625~~	~~0~~	~~0~~	~~2559.9609375~~

C.2.46 DID 0x0245-intake manifold pressure (absolute)

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
intake manifold pressure (absolute)	2	hPa	0.0390625	0	0	2559.960938

C.2.47 DID 0x0246-Number of clutch operations detected

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Number of clutch operations detected	1	–	1	0	0	255

C.2.48 DID 0x0247-intake air temperature, linearised and calculated

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
intake air temperature, linearised and calculated	1	℃	0.75	-48	-48	143.25

C.2.49 DID 0x0248-vehicle speed output value to scan tool

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
vehicle speed output value to scan tool	2	km/h	0.0078125	0	0	511.9921875

C.2.50 DID 0x0249-battery voltage; scanned value of ADC

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
battery voltage; scanned value of ADC	2	V	0.004882813	0	0	319.9951172

C.2.51 DID 0x0252-Angle: intake valve opens related to LWOT

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Angle: intake valve opens related to LWOT	2	KW	0.0078125	0	0	511.9921875

C.2.52 DID 0x0254- desired camshaft angle inlet valve opened

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
desired camshaft angle inlet valve opened	2	KW	0.0078125	0	0	511.9921875

C.2.53 DID 0x0256-oil temperature

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
oil temperature	2	℃	0.0234375	-273.15	-273.15	1262.826563

C.2.54 DID 0x0257-lambda controller output

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
lambda controller output	2	–	3.0518E-05	0	0	1.999969482

C.2.55 DID 0x0258-multiplicative correction of the mixture adaptation

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
multiplicative correction of the mixture adaptation	2	–	3.0518E-05	0	0	1.999969482

~~C.2.56 DID 0x0259-Lambda controller output bank 2~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~Lambda controller output bank 2~~	~~2~~	~~–~~	~~3.0518E-05~~	~~0~~	~~0~~	~~1.999969482~~

~~C.2.57 DID 0x025A- multiplicative correction of the mixture adaptation bank 2~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~multiplicative correction of the mixture adaptation bank 2~~	~~2~~	~~–~~	~~3.0518E-05~~	~~0~~	~~0~~	~~1.999969482~~

C.2.58 DID 0x025B-sensor voltage from throttle potentiometer 1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
sensor voltage from throttle potentiometer 1	2	V	0.001220703	0	0	79.9987793

C.2.59 DID 0x025C-sensor voltage from throttle potentiometer 2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
sensor voltage from throttle potentiometer 2	2	V	0.001220703	0	0	79.9987793

C.2.60 DID 0x0260-additive correction of the mixture adaptation

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
additive correction of the mixture adaptation	2	-	0.046875	-1536	-1536	1535.953125

C.2.61 DID 0x0262-Voltage PWG potentiometer 1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Voltage PWG potentiometer 1	2	V	0.004882813	0	0	319.9951172

C.2.62 DID 0x0263-Voltage PWG potentiometer 2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Voltage PWG potentiometer 2	2	V	0.004882813	0	0	319.9951172

C.2.63 DID 0x0264-Doubled PWG potentiometer-2 voltage

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Doubled PWG potentiometer-2 voltage	2	V	0.004882813	0	0	319.9951172

C.2.64 DID 0x0265-Actual value internal res. Ri-Nernst cell lambda sensor downstream cat

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Actual value internal res. Ri-Nernst cell lambda sensor downstream cat	2	Ohm	2	0	0	131070

C.2.65 DID 0x0266-Exhaust gas temperature in front of precatalyst out of model

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Exhaust gas temperature in front of precatalyst out of model	2	℃	0.0234375	-273.15	-273.15	1262.826563

~~C.2.66 DID 0x0267-HFM air mass value~~

~~Data Name~~	~~#of bytes~~	~~Unit~~	~~Scaling~~	~~Offset~~	~~Min.~~	~~Max.~~
~~HFM air mass value~~	~~2~~	~~kg/h~~	~~0.1~~	~~0~~	~~0~~	~~6553.5~~

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C.2.67 DID 0x0268-Transfer function (Mrad/Mkurbelwelle) from transmission control

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Transfer function (Mrad/Mkurbelwelle) from transmission control	1	–	0.1	0	0	25.5

C.2.68 DID 0x026B-Interface for mode $01+$02 short term fuel trim bank 1 for PID $06

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Interface for mode $01+$02 short term fuel trim bank 1 for PID $06	1	-	0.78125	0	0	199.21875

C.2.69 DID 0x026C- Interface for mode $01+$02 long term fuel trim bank 1 for PID $07

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Interface for mode $01+$02 long term fuel trim bank 1 for PID $07	1	–	0.78125	0	0	199.21875

C.2.70 DID 0x026D-injection time

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
injection time	2	ms	0.001	0	0	4294967.295

C.2.71 DID 0x026E-effective steady-state idle speed

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
effective steady-state idle speed	1	rpm	10	0	0	2550

C.2.72 DID 0x0278-Location of oxygen sensors

To be defined by ECM supplier and confirmed by SGMW.

C.2.73 DID 0x027A- OBD requirements to which vehicle is designed

Data Name	#of bytes	Description
OBD requirements to which vehicle is designed	1	Refer to the definition of “OBD requirements to which vehicle is designed” in the document of 15031-5.

C.2.74 DID 0x027B-counter for distance travelled while MIL is activated

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
counter for distance travelled while MIL is activated	2	km	1	0	0	65535

C.2.75 DID 0x027C-engine temperature at start

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
engine temperature at start	1	℃	0.75	-48	-48	143.25

C.2.76 DID 0x027D-standardized mass flow from canister purge control

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
standardized mass flow from canister purge control	2	kg/h	0.000390625	0	0	25.59960938

C.2.77 DID 0x027E-basic ignition angle

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
basic ignition angle	1	Grad KW	0.75	0	0	191.25

C.2.78 DID 0x027F-fault counter, summary, counts emission relevant misfirings of all cylinders

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
fault counter, summary, counts emission relevant misfirings of all cylinders	2	–	1	0	0	65535

C.2.79 DID 0x0284-area of misfire, minimum engine speed

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
area of misfire, minimum engine speed	2	rpm	0.25	0	0	16383

C.2.80 DID 0x0285-area of misfire, maximum engine speed

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
area of misfire, maximum engine speed	2	rpm	0.25	0	0	16383

C.2.81 DID 0x0286-area of misfirings, minimum load

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
area of misfirings, minimum load	1	%	0.390625	0	0	99.609375

C.2.82 DID 0x0287-area of misfirings, maximum load

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
area of misfirings, maximum load	1	%	0.390625	0	0	99.609375

C.2.83 DID 0x0288-Cycle time forced lambda modulation

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Cycle time forced lambda modulation	2	–	1	0	0	65535

C.2.84 DID 0x0289-desired torque change from the idle speed control (PD-part)

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
desired torque change from the idle speed control (PD-part)	2	%	0.003051758	-100	-100	99.99694824

C.2.85 DID 0x02A2-Actual intake manifold pressure sensor voltage

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Actual intake manifold pressure sensor voltage	2	V	0.004882813	0	0	319.9951172

C.2.86 DID 0x02AA-O_2 sensor voltage bank1, sensor 1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
O_2 sensor voltage bank1, sensor 1	2	V	0.004882813	0	0	319.9951172

C.2.87 DID 0x02AB-O_2 sensor voltage bank1, sensor 2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
O_2 sensor voltage bank1, sensor 2	2	V	0.004882813	0	0	319.9951172

C.2.88 DID 0x0311-ignition-timing retardation with retardation for dynamics, Cylinder #1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ignition-timing retardation with retardation for dynamics, Cylinder #1	1	Grad KW	0.75	0	0	191.25

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C.2.89 DID 0x0312-ignition-timing retardation with retardation for dynamics, Cylinder #2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ignition-timing retardation with retardation for dynamics, Cylinder #2	1	Grad KW	0.75	0	0	191.25

C.2.90 DID 0x0313-ignition-timing retardation with retardation for dynamics, Cylinder #3

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ignition-timing retardation with retardation for dynamics, Cylinder #3	1	Grad KW	0.75	0	0	191.25

C.2.91 DID 0x0314-ignition-timing retardation with retardation for dynamics, Cylinder #4

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ignition-timing retardation with retardation for dynamics, Cylinder #4	1	Grad KW	0.75	0	0	191.25

C.2.92 DID 0x0321-reference level knock control, 16bit, Cylinder #1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
reference level knock control, 16bit, Cylinder #1	2	V	7.6294E-05	0	0	4.999923706

C.2.93 DID 0x0322-reference level knock control, 16bit, Cylinder #2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
reference level knock control, 16bit, Cylinder #2	2	V	7.6294E-05	0	0	4.999923706

C.2.94 DID 0x0323-reference level knock control, 16bit, Cylinder #3

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
reference level knock control, 16bit, Cylinder #3	2	V	7.6294E-05	0	0	4.999923706

C.2.95 DID 0x0324-reference level knock control, 16bit, Cylinder #4

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
reference level knock control, 16bit, Cylinder #4	2	V	7.6294E-05	0	0	4.999923706

C.2.96 DID 0x0331-number of misfire: cylinder#1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
number of misfire: cylinder#1	2	–	1	0	0	65535

C.2.97 DID 0x0332-number of misfire: cylinder#2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
number of misfire: cylinder#2	2	–	1	0	0	65535

C.2.98 DID 0x0333-number of misfire: cylinder#3

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
number of misfire: cylinder#3	2	–	1	0	0	65535

C.2.99 DID 0x0334-number of misfire: cylinder#4

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
number of misfire: cylinder#4	2	–	1	0	0	65535

C.2.100 DID 0x0335-Cycle time forced lambda modulation

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Cycle time forced lambda modulation	2	–	1	0	0	65535

C.2.101 DID 0x0D00-battery voltage

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
battery voltage	2	V	0.000390625	0	0	25.59960938

C.2.102 DID 0x0D01-vehicle speed

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
vehicle speed	2	km/h	0.0078125	0	0	511.99218

C.2.103 DID 0x0D05-indicated real engine torque

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
indicated real engine torque	2	%	0.001525879	0	0	99.99847

C.2.104 DID 0x0D06-Drivingdistance since prime start

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Drivingdistance since prime start	4	km	1	0	0	4294967295

C.2.105 DID 0x0F00-Coolant temperature of engine misses

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Coolant temperature of engine misses	1	℃	0.75	-40	0	151.25

C.2.106 DID 0x0F01-Duty cycle for carbon

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Duty Cycle for Carbon	1	%	0.390625	0	0	99.609375

C.2.107 DID 0x0F02-Intake VVT control PWM

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Intake VVT Control PWM	1	%	0.390625	0	0	99.609375

C.2.108 DID 0x0F03-Upstream O2 heater status

Byte No.	Description	Data Type	Value (Hex)
data #1	Upstream O_2 heater status	Hex	0x01：heater ON 0x00：heater OFF

C.2.109 DID 0x0F04-Downstream O2 heater status

Byte No.	Description	Data Type	Value (Hex)
data #1	Downstream O_2 heater status	Hex	0x01：heater ON 0x00：heater OFF

C.2.110 DID 0x0F05-Injection time, cylinder#1

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Injection time cylinder#1	2	ms	0.002667	0	0	174.76

C.2.111 DID 0x0F06-Injection time, cylinder#2

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Injection time cylinder#2	2	ms	0.002667	0	0	174.76

C.2.112 DID 0x0F07-Injection time, cylinder#3

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Injection time cylinder#3	2	ms	0.002667	0	0	174.76

C.2.113 DID 0x0F08-Injection time, cylinder#4

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Injection time cylinder#4	2	ms	0.002667	0	0	174.76

C.2.114 DID 0x0F09-λ closed loop control status

Byte No.	Description	Data Type	Value (Hex)
data #1	λ closed loop control status	Hex	0x01：ON 0x00：OFF

C.2.115 DID 0x0F0A-Fuel pump relay status

Byte No.	Description	Data Type	Value (Hex)
data #1	Fuel pump relay status	Hex	0x01：ON 0x00：OFF

C.2.116 DID 0x0F0B-Main relay status

Byte No.	Description	Data Type	Value (Hex)
data #1	Main relay status	Hex	0x01：ON 0x00：OFF

C.2.117 DID 0x0F0C-Cooling fan 2 switch status

Byte No.	Description	Data Type	Value (Hex)
data #1	Cooling fan 2 switch status	Hex	0x01：ON 0x00：OFF

C.2.118 DID 0x0F0D-Cooling fan 1 switch status

Byte No.	Description	Data Type	Value (Hex)
data #1	Cooling fan 1 switch status	Hex	0x01：ON 0x00：OFF

C.2.119 DID 0x0F0E-A/C compressor clutch relay status

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Byte No.	Description	Data Type	Value (Hex)
data #1	A/C compressor clutch relay status	Hex	0x01：ON 0x00：OFF

C.2.120 DID 0x0F0F-A/C request switch status

Byte No.	Description	Data Type	Value (Hex)
data #1	A/C request switch status	Hex	0x01：ON 0x00：OFF

C.2.121 DID 0x0F10-A/C evaporator pressure

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
A/C evaporator pressure	1	hpa	156.86	0	0	39999.3

C.2.122 DID 0x0F11-A/C evaporator temperature

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
A/C evaporator temperature	1	℃	0.7501875468867217	48.01200300075019	0	143.2858214553638

C.2.123 DID 0x0F12-Break test switch status

Byte No.	Description	Data Type	Value (Hex)
data #1	Break test switch status	Hex	0x01：ON 0x00：OFF

C.2.124 DID 0x0F13-Break lamp switch status

Byte No.	Description	Data Type	Value (Hex)
data #1	Break lamp switch status	Hex	0x01：ON 0x00：OFF

C.2.125 DID 0x0F14-Clutch switch status

Byte No.	Description	Data Type	Value (Hex)
data #1	Clutch switch status	Hex	0x01：ON 0x00：OFF

C.2.126 DID 0x0F15-MIL indicator

Byte No.	Description	Data Type	Value (Hex)
data #1	MIL indicator	Hex	0x01：ON 0x00：OFF

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C.2.127 DID 0x0F16-Cruise status

Byte No.	Description	Data Type	Value (Hex)
data #1	Cruise status	Hex	0x01：Cruise ON 0x00：Cruise OFF

C.2.128 DID 0x0F17-Cruise speed

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Cruise speed	2	Km/h	1	1	0	65535

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Annex D (normative) Stored data transmission functional unit data parameter definitions

The detailed description about this Annex can be found in Annex D of the document *CN200S General Diagnostic Requirement Specification*.

It is possible for ECM supplier to add the additional definitions in Annex D for CN200S vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex E (normative) Input output control functional unit data parameter definitions

E.1 InputOutputControlParameter definitions

The detailed description about this Annex can be found in Annex E.1 of the document *CN200S General Diagnostic Requirement Specification*.

It is possible for ECM supplier to add the additional definitions in Annex E.1 for CN200S vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

E.2 InputOutputControlByIdentifier (2F hex) controlOptionRecord definitions

The following table lists the definition about InputOutput ControlState.

E.3InputOutputControlByIdentifier (2F hex) controlEnableMaskRecorddefinitions

N/A

E.4InputOutputControlByIdentifier (2F hex) controlStatusRecorddefinitions

The definition of controlStatusRecordis the same as controlOptionRecord which defined in chapter E.2 of Annex E.

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Annex F (normative) Remote activation of routine functional unit data parameter definitions

F.1 RoutineIdentifier definition

The detailed description about this Annex can be found in Annex F.1 of the document *CN200S General Diagnostic Requirement Specification*.

It is possible for ECM supplier to add the additional definitions in Annex F.1 for CN200S vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

F.2 RoutineControl (31 hex) routineControlOption definitions

The following table lists the definition about routine control DID.

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Annex G (informative) Examples for addressAndLengthFormatIdentifier parameter values

The detailed description about this Annex can be found in Annex G of the document *CN200S General Diagnostic Requirement Specification*.

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